3.1 Physical chemistry

3.1.1 Atomic structure

The chemical properties of elements depend on their atomic structure and in particular on the arrangement of electrons around the nucleus. The arrangement of electrons in orbitals is linked to the way in which elements are organised in the Periodic Table. Chemists can measure the mass of atoms and molecules to a high degree of accuracy in a mass spectrometer. The principles of operation of a modern mass spectrometer are studied.

3.1.1.1 Fundamental particles

Content	Opportunities for skills development
Appreciate that knowledge and understanding of atomic structure has evolved over time. Protons, neutrons and electrons: relative charge and relative mass. An atom consists of a nucleus containing protons and neutrons surrounded by electrons.	

3.1.1.2 Mass number and isotopes

Content	Opportunities for skills development
Mass number (*A*) and atomic (proton) number (*Z*). **Students should be able to:** • determine the number of fundamental particles in atoms and ions using mass number, atomic number and charge • explain the existence of isotopes. The principles of a simple time of flight (TOF) mass spectrometer, limited to ionisation, acceleration to give all ions constant kinetic energy, ion drift, ion detection, data analysis. The mass spectrometer gives accurate information about relative isotopic mass and also about the relative abundance of isotopes. Mass spectrometry can be used to identify elements. Mass spectrometry can be used to determine relative molecular mass. **Students should be able to:** • interpret simple mass spectra of elements • calculate relative atomic mass from isotopic abundance, limited to mononuclear ions.	**MS 1.1** Students report calculations to an appropriate number of significant figures, given raw data quoted to varying numbers of significant figures. **MS 1.2** Students calculate weighted means, eg calculation of an atomic mass based on supplied isotopic abundances. **MS 3.1** Students interpret and analyse spectra.

3.1.1.3 Electron configuration

Content	Opportunities for skills development
Electron configurations of atoms and ions up to *Z* = 36 in terms of shells and sub-shells (orbitals) s, p and d. Ionisation energies. **Students should be able to:** • define first ionisation energy • write equations for first and successive ionisation energies • explain how first and successive ionisation energies in Period 3 (Na–Ar) and in Group 2 (Be–Ba) give evidence for electron configuration in sub-shells and in shells.	